

Professional Summary

Lead AI research engineer driving the core generative modeling behind **VARCO 3D**, NC AI's production 3D asset generation service with **30K+ users** and **300K+ generated assets**. Own research direction and deployed model behavior across native 3D representations, large-scale generative training, inference optimization, and production mesh delivery. Specialized in **3D generative modeling** and **GPU systems**, spanning diffusion and flow models, CUDA, mesh processing, and neural rendering. Published at **ICML** and **CVPR**, with invited talks at KAIST and Korean government institutions.

Core Expertise

3D Generative Models	Native 3D representations, latent diffusion, flow matching, DiT architectures, geometry generation, texture synthesis
GPU / ML Systems	PyTorch, CUDA, distributed training, inference optimization, profiling, deterministic numerical validation, data pipeline design
Geometry / Graphics	Mesh processing, SDF/UDF, sparse voxels, textured meshes, Blender, Three.js, real-time rendering workflows
Neural Rendering	Neural Radiance Fields, Gaussian splatting, differentiable rendering, real-time engine integration

Experience

Lead AI Research Engineer Feb 2025 – Present
NC AI, VARCO 3D Team

- Lead core generative modeling for **VARCO 3D**, owning model architecture, training, evaluation, inference optimization, and production deployment.
- Scaled 3D generative model training and inference to the multi-billion-parameter regime across 64+ A100/H100 GPUs while meeting production cost and latency constraints.
- Improved geometric fidelity, reducing Chamfer Distance by 23%, while cutting inference latency by 74% against the previous internal baseline through architectural and inference-time redesign.
- Lead research across native 3D representations, large-scale diffusion and flow training, inference-time controllability, and GPU-accelerated generation pipelines.

AI Researcher Mar 2023 – Feb 2025
NCSOFT AI Lab, 3D Generation & Neural Rendering Team (now NC AI)

- Researched 3D asset generation, native 3D latent representations, and texture synthesis for game-content production workflows; this research line led to **CaPa**, a production-oriented 4K textured mesh generation pipeline.
- Developed neural-rendering workflows for real-time engine integration and interactive 3D scene representation, demonstrated in NCSOFT's AI-based 3D rendering project.

AI Researcher, Intern Jul 2022 – Dec 2022
NAVER AI Lab, Mentor: Jin-Hwa Kim, Ph.D.

- Researched robust camera pose refinement for multi-resolution hash encoding in NeRF, addressing reconstruction instability under noisy poses; published at **ICML 2023**.

Selected Work

VARCO 3D: Production 3D Asset Generation Jul 2024 – Present
NC AI | Service | Case Study

- Lead the core generative system behind NC AI's production 3D asset generation service, spanning native 3D representation design, large-scale training, inference optimization, and production mesh delivery.

CaPa: Efficient 4K Textured Mesh Generation 2024
NCSOFT Research | Project Page

- Developed a production-oriented textured mesh generation pipeline combining geometry carving and high-resolution texture

synthesis for efficient 4K 3D asset creation.

Selected Publications & Technical Reports

CaPa: Carve-n-Paint Synthesis for Efficient 4K Textured Mesh Generation

H. Heo, J. Kim, S. Lee, J. Wi, J. Choi, S. Ahn*

Technical Report | [Link](#)

Efficient high-resolution textured mesh generation pipeline for production-oriented 3D asset creation.

Robust Camera Pose Refinement for Multi-Resolution Hash Encoding

H. Heo, T. Kim, J. Lee, J. Lee, S. Kim, H. J. Kim*, J.-H. Kim*

ICML 2023 | [Link](#)

Camera pose refinement for NeRF-style multi-resolution hash encoding under pose noise and reconstruction instability.

Semantic-aware Occlusion Filtering Neural Radiance Fields in the Wild

J. Lee, I. Kim, H. Heo, H. J. Kim*

Preprint | [Link](#)

Semantic filtering for more robust in-the-wild neural radiance field reconstruction.

Consistency Learning via Decoding Path Augmentation for Transformers in Human Object Interaction Detection

J. Park, S. Lee, H. Heo, H. Choi, H. J. Kim*

CVPR 2022 | [Link](#)

Consistency learning for transformer-based human-object interaction detection through decoding-path augmentation.

Temporally Sparse Adversarial Perturbations on Videos

H. Heo, D. Ko, J. Lee, Y. Hong*, H. J. Kim*

IEEE Access 2021 | [Link](#)

Video adversarial perturbation method with temporally sparse attack structure.

Education

M.S. in Computer Science & Engineering

Sep 2020 – Feb 2023

Korea University, Seoul | Advisor: Prof. Hyunwoo J. Kim, MLV Lab

GPA: 4.3/4.5 · Publications: ICML 2023, CVPR 2022, IEEE Access 2021

B.S. in Computer Science & Engineering

Mar 2015 – Aug 2020

Korea University, Seoul

Professional Activities

Invited Talks

- *An Introduction to 3D Generative AI* — KAIST, Mar 2026
- *Spatial 3D AI Technology and Its Applications* — Ministry of Land, Infrastructure and Transport, Korea, Sep 2025
- *Radiance Fields and Spatial 3D AI for Digital Twins* — National Geographic Information Institute, Korea, Aug 2025
- *Rasterization vs. Ray Tracing with 3D Gaussian Primitives* — NCSOFT Developer Seminar, Sep 2024
- *Joint Optimization of Camera Pose and Accelerated Radiance Fields* — NAVER Cloud, Mar 2023